INCH-POUND
MIL-DTL-32777
18 October 2024
SUPERSEDING
A-A-59529
13 December 1999

DETAIL SPECIFICATION

CAP, FOOD HANDLER'S

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This detailed specification covers the requirements for a food handler's cap (see 1.2 and 6.2).

1.2 Classification. The cap is available in one type and includes the following sizes.

1.2.1 Schedule of sizes.

Small (6-3/4)
Medium (7-1/8)
Large (7-1/2)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 8415

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract (see 6.2).

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199	- Thread, Polyester Core, Cotton or Polyester-Covered
A-A-52094	- Thread, Cotton

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075	- Label: For Clothing, Equipage, and Tentage (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of the document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Evaluation Procedures

AATCC EP1	- Gray Scale for Color Change
AATCC EP2	- Gray Scale for Staining
AATCC EP8	- 9-Step Chromatic Transference Scale

Test Methods

AATCC TM8	- Colorfastness to Crocking: Crockmeter Method
AATCC TM15	- Colorfastness to Perspiration
AATCC TM16.3	- Colorfastness to Light: Xenon-Arc
AATCC TM20A	- Fiber Analysis: Quantitative
AATCC TM61	- Colorfastness to Laundering: Accelerated
AATCC TM81	- pH of the Water-Extract from Wet Processed Textiles
AATCC TM124	- Smoothness Appearance of Fabrics after Repeated Home Laundering
AATCC TM130	- Soil Release: Oily Stain Release Method
AATCC TM135	- Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D1424	- Standard Test Method for Tearing Strength of Fabric by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D1776/D1776M	- Standard Practice for Conditioning and Testing Textiles
ASTM D3775	- Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3990	- Standard Terminology Relating to Fabric Defects
ASTM D4032	- Standard Test method for Stiffness of Fabric by the Circular Bend Procedure
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193	- Standard Practices for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample caps for the first article inspection and approval (see 4.2 and 6.2).

3.2 Design. The cap shall be of a turn-up style, two-piece construction, consisting of a two-ply circular headband and a flat circular crown. When worn, the headband shall be turned up to form four plies (see Figure 1).

3.3 Figures. Figures 1, 2, and 3 are furnished for informational purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.4. Materials. The cap materials shall conform to the applicable specifications and standards as listed or required herein.

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3.4.1 Basic material. The cloth used for the headband of the cap shall be polyester/cotton twill material conforming to the requirements specified in Table I.

TABLE I. Physical requirements of basic material

Characteristic	Requirement
Fiber content	65 (± 5) % polyester the remainder cotton
Weave	2/1 left hand twill
Yarns per inch (minimum) Warp Filling	 124 54
Weight, ounces per square yard (minimum)	6.0
Breaking strength, pounds (minimum) Warp Filling	 165 65
Tear strength, pounds (minimum) Warp Filling	 7.0 5.0
pH	5.5 – 8.5
Dimensional change, percent (maximum) Warp Filling	 3.0 3.0
Appearance after 5 launderings (minimum)	SA-3
Stiffness, psi (maximum)	1
Soil release (minimum)	Grade 3.5
Discoloration (minimum) Laundering Color Change Staining Perspiration Acid Color Change Staining Alkaline Color Change Staining Light Crocking Dry Wet	 3 3 3 3 3 3 3 3 3

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3.4.2 Net cloth. The cloth used for the crown of the cap shall be polyester raschel knit conforming to the requirements specified in Table II.

TABLE II. Physical requirements of net cloth

Characteristic	Requirements
Meshes per inch (minimum) Warp Filling	 5 5
Weight, ounces per square yard	1.8 – 4.0
pH	5.5 – 8.5

3.4.3 Thread. The thread for seaming and stitching the cap shall be Type I of A-A-50199, 2 ply cotton-covered polyester-core, Tex size 36-45 (needle and bobbin) and 31-35 (looper) and shall approximate the color of the basic cloth. As an alternate, the thread may be Type I of A-A-52094, 3 ply cotton Tex size 60 (needle and bobbin) and Tex size 40 (looper) approximating the color of the basic cloth.

3.4.4 Color. The color of the cloth materials and components shall be white.

3.4.5 Labels. Each cap shall have a combination size, identification, and instruction label, and a barcode label.

3.4.5.1 Combination size, identification, and instruction label. Each cap shall have a combination size, identification, personal and care label. The size, identification, personal and care label shall conform to Type VI, Class 14 of MIL-DTL-32075 when tested as specified in 4.4.2. Label color shall be white. The label shall be caught in the headband and crown joining seam. The combination label shall be facing the head and shall not be visible from the outside when the cap is in use. The combination size, identification, and instruction label shall include the following information and be in the following configuration:

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Medium (7-1/8) (EXAMPLE)
NSN: 8415-00-000-0000 (EXAMPLE)
CAP, FOOD HANDLER'S
65% Polyester, 35% Cotton (EXAMPLE)
CONTRACT NO.: SPE1C1-00-0-0000 (EXAMPLE)
CONTRACTOR'S NAME:

LAUNDERING INSTRUCTIONS:
Machine wash in hot water (140 °F Max)
Or
Hand wash using mild detergent, rinse completely.
DO NOT STARCH.
Tumble dry medium heat and remove immediately from dryer.
DO NOT WRING OR TWIST.
Light ironing **if needed.**
DO NOT COMMERCIALLY HOT PRESS.
Or
FOLLOW SHIPBOARD WASH FORMULA II
DO NOT REMOVE THIS LABEL

3.4.5.3 Label, barcode. Each cap shall be individually bar-coded in accordance with MIL-DTL-32075, Type VIII, Class 17. The tags shall have a hole and shall be attached to each cap by a fastener. The bar-coding element shall be legible and readable by scanner. The bar code element shall be a 13-digit national stock number (NSN). There shall be a 12-digit Universal Product Code (UPC) assigned for all NSNs by the government. The initial "UPC" shall appear beneath the code. The bar codes for the NSN and UPC shall be medium to high density and shall be located so that they are completely visible on the item when it is folded and/or packaged as specified. The label shall not cause any damage to the cap such as a spot or stain or damage due to the location on the item.

3.4.6 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, or environmentally preferable or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Construction.

3.5.1 Cutting. The cap's cloth shall be cut in strict accordance with the government furnished patterns.

3.5.2 Patterns. The Government will furnish a complete set of patterns which show size, grain lines, cut quantity, placement markings and notches for assembly. The Government's patterns shall be used to create the contractor's working patterns. Minor modifications are permitted to the contractor's working pattern where necessary when using automatic equipment or production

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processes. These modifications shall not alter the appearance, serviceability, or dimensional requirements cited in this specification. Patterns provide a seam allowance of 1/4-inch which shall be maintained throughout.

3.5.2.1 Pattern parts. The cap shall be manufactured from the pattern parts below.

TABLE III. Pattern parts

Material	CAD nomenclature	Specification nomenclature	Cut parts
Basic material	PD59529-HEADBAND	Headband	1
Net cloth	PD59529-CROWN	Crown	1

3.6 Manufacturing.

3.6.1 Stitches, seams and stitching. The stitches, seams, and stitching types specified in Table IV shall conform to ASTM D6193. The manufacturer is not required to follow the seam type sequence cited under Table IV for the construction of the cap; however, the finished cap construction shall conform to requirements cited under this specification.

TABLE IV. Stiches, seams and type

Seam Placement	Seam Type	Gauge	Stitch Type
Headband construction	SSa-1	1/4 from edge	301 or 401
Overedge joining of headband and crown	SSa-1	1/4 from edge	Any 500 series
Edge stitch headband	LSq-2	1/16 to 1/8 inch from edge	301

3.7.2 Seams. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching spacing, and parallelism shall be maintained with a tolerance of ±1/32 inch. Seam construction shall be maintained with seams sewn so that no run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any may be used.

3.7.3 Stitching. The seams shall be sewn with 12–14 stitches per inch.

3.7.3.1 Trimming. All thread ends shall be trimmed to a length of not more than 1/4-inch.

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3.7.3.2 Types 301 stitching. Ends of all 301 stitching shall be backstitched or overstitched not less than 1/4-inch except when caught in other seams or stitching. Ends of a continuous line of stitching shall overlap no less than 1/2-inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn or resulting in breaking of stitches when normal pull is applied.

3.7.3.3 Type 301 repair. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin run outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/4-inch back of the end of the stitching. When making the above repairs, the ends of stitching are not required to be backstitched.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/4-inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and re-stitching in the required manner. When making the above repairs, the ends of stitching are not required to be backstitched.

3.8 Finished dimensions. The finished cap shall conform to the finished measurements and tolerances specified in Table V as indicated in section 4.6.

TABLE V. Finished measurements (inches)

Size	Inside circumference (A)	Height of headband (B)
Small (6-3/4)	21-1/4 (–0, +3/8)	7-1/2 (± 1/4)
Medium (7-1/8)	22-3/8 (–0, +3/8)	7-1/2 (± 1/4)
Large (7-1/2)	23-1/2 (–0, +3/8)	7-1/2 (± 1/4)

3.9 Workmanship. The finished cap shall be uniform in quality and free from defects that adversely affect form, fit, or function, and those specified in Table VII, and defects that can be identified in accordance with ASTM D3990.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. The first article inspection of the cap shall be performed on finished end items for all the requirements in Section 3 and shall be inspected as specified in 4.5 and 4.6 for compliance with design, construction, workmanship and dimensional requirements.

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4.2.1 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2), the number of samples for first article inspections shall be as specified in the procurement document. The sample unit shall be one cap. The lot size shall be expressed in units of caps.

4.3 Conformance inspection. Conformance inspection shall consist of all physical properties testing in Table VI, and examination for defects in Table VII.

4.3.1 Conformance inspection sample. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4. The sample shall be one finished cap. The lot size shall be expressed in units of caps.

4.4 Inspection methods.

4.4.1 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with requirements specified in Section 3. Materials and components, which can be classified as having a defect in accordance with Table VII, shall be removed from production.

4.4.2 Component and material verification. In accordance with 4.3, components shall be examined and tested in accordance with all the requirements in Table VI. The components shall be conditioned as directed in ASTM D1776/D1776M. All test reports shall contain the individual values utilized in expressing the results.

TABLE VI. Component and material verification

Material	Characteristic	Requirement	Verification
Basic material	Fiber content	3.4.1	AATCC TM20A
	Weave	3.4.1	Visual
	Yarns per inch (minimum) Warp Filling	 3.4.1 3.4.1	ASTM D3775
	Weight, ounces per square yard (minimum)	3.4.1	ASTM D3776/D3776M, Option C
	Breaking strength, pounds (minimum) Warp Filling	 3.4.1 3.4.1	ASTM D5034
	Tear strength, pounds (minimum) Warp Filling	 3.4.1 3.4.1	ASTM D1424
	pH	3.4.1	AATCC TM81

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TABLE VI. Component and material verification - Continued

Material	Characteristic	Requirement	Verification
	Dimensional change, percent (maximum) Warp Filling	 3.4.1 3.4.1	AATCC TM135 (1) V Ai
	Appearance after 5 launderings (minimum)	3.4.1	AATCC TM124 (1) V Ai
	Stiffness, psi (maximum)	3.4.1	ASTM D4032
	Soil release (minimum)	3.4.1	AATCC TM130 1/
	Discoloration to: (minimum) Laundering (3 cycles) Color Change Staining Perspiration Acid Color Change Staining Alkaline Color Change Staining Light Crocking Dry Wet	 3.4.1 3.4.1 3.4.1 3.4.1 3.4.1 3.4.1 3.4.1 3.4.1 3.4.1	 AATCC TM61, Test 2A 2/, 3/ AATCC TM15 2/ 3/ AATCC TM16.3 2/ 4/ AATCC TM8 5/
Net cloth	Meshes per Inch Warp Filling	3.4.2	ASTM D3775
	Weight, ounces per square yard	3.4.2	ASTM D3776/D3776M
	pH	3.4.2	AATCC TM81
Thread	All	3.4.3	A-A-50199 6/
Label	All	3.4.5.1	MIL-DTL-32075

1/ Tested using washing procedure V (60 ± 3°C (140 ± 5°F)) and evaluated after 1 wash cycle.
2/ Rated using the AATCC EP1, Gray Scale for Color Change.
3/ Rated using the AATCC EP2, Gray Scale for Staining. Only the stain on the polyester and cotton fibers of the multifiber transfer swatch shall be evaluated.
4/ The samples shall be exposed to Option 3 for 170 kJ at 420 nm or after 40 AFU.
5/ Rated using the AATCC EP8, AATCC 9-Step Chromatic Transference Scale. The black crock cloth shall be used.
6/ A-A 52094 shall be used for alternate thread.

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4.5 End item visual examination. Each finished cap selected for visual examination shall be examined for defects in color, design, material, construction, workmanship, and marking and the defects classified in accordance with Table VII. ASTM D3990 shall be used to aid in identifying defects when needed. Any defect that is not identified in Table VII but is on a finished cap and is cited under ASTM D3990 or MIL-STD-1491 shall be classified as a minor defect.

TABLE VII. Visual examination

Examination	Defect	Classification	
		Major	Minor
Material	Hole, cut, tear, smash, burn, drill hole, run, needle chew, visible mend, thin place, misweave affecting appearance or serviceability.	101	
	Material not as specified	102	
Components and assembly	Any component part not as specified; operation improperly performed.	103	
	Any component omitted.	104	
	Any component dimensions not as specified.	105	
Seams and stitching	Seams puckered, distorted, pleated, wavy, twisted.		201
	Seams irregular or open seam, raw edge affecting appearance or serviceability.	106	
	Seam, seam allowance, or stitch type not as specified.		202
Cleanliness	Spot, stain, excessive thread ends not trimmed or removed, odor.	107	
Labels	Any label omitted, incorrect type, size or configuration, illegible, inaccurate, improperly attached as specified.	108	
	Human-readable interpretation (HRI) omitted, or illegible		203
	Bar code label not visible on packaged item		204
	Bar code attachment causes damage to the item	109	

4.6 End item dimensional examination. Each finished cap selected for dimensional examination shall be measured in accordance with Table VIII. Any dimension not within the specified tolerance shall be classified as a major defect.

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TABLE VIII. Dimensional Examination

Measurement	Requirement	Description
Inside Circumference (A)	3.8	With the headband unfolded, measure the inside circumference of the headband as indicated in Figure 2
Height of Headband (B)	3.8	With the headband unfolded, measure from the base of the headband to top of the headband as indicated in Figure 3

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cap covered by this specification is intended for use by cooks and bakers of the Department of the Navy.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specific issue of individual documents referenced (see 2.2 and 2.3)
c. Quantity (see 1.2)
d. National Stock Number.
e. Applicable Government patterns, including revisions.
f. When first article is required (see 3.1).
g. Number of first article inspection samples (see 4.2.1)
h. Name and address of the first inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.
i. Inspection conditions.
j. Conformance inspection and acceptable quality limits (see 4.3)
k. Packaging required (see 5.1)

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6.3 Information requests. For access to information such as patterns go to Defense Logistics Agency Troop Support's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military services.

6.4 Subject term (key word) listing.

Food service
Cooks
Bakers

Custodians:
Navy – NU
DLA – CT

Preparing Activity:
DLA – CT
(Project 8415-2024-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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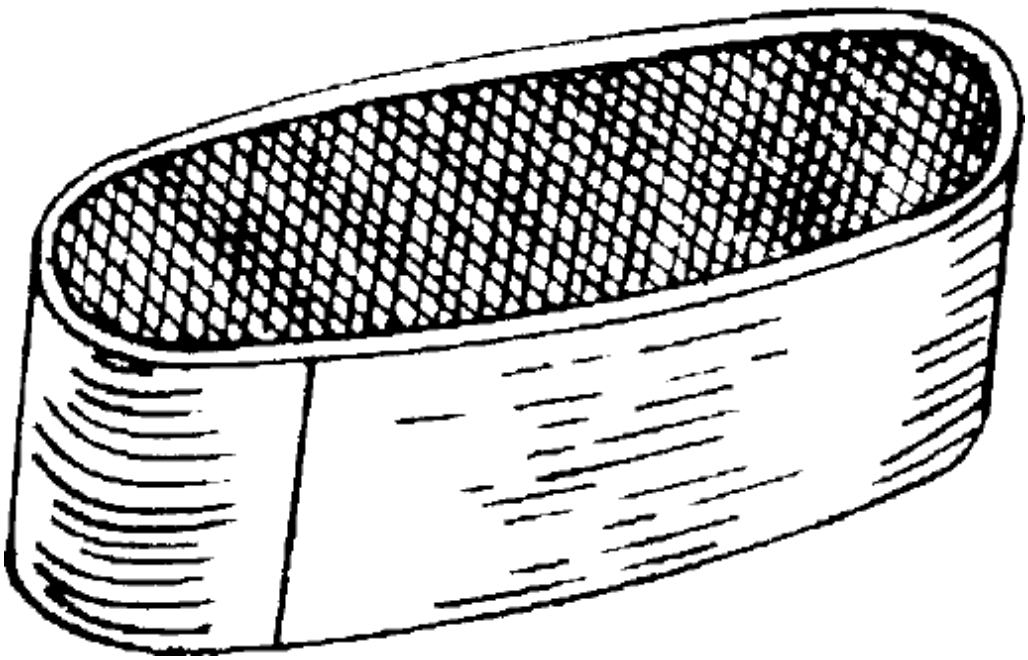


FIGURE 1. Folded - Food Handler’s Cap

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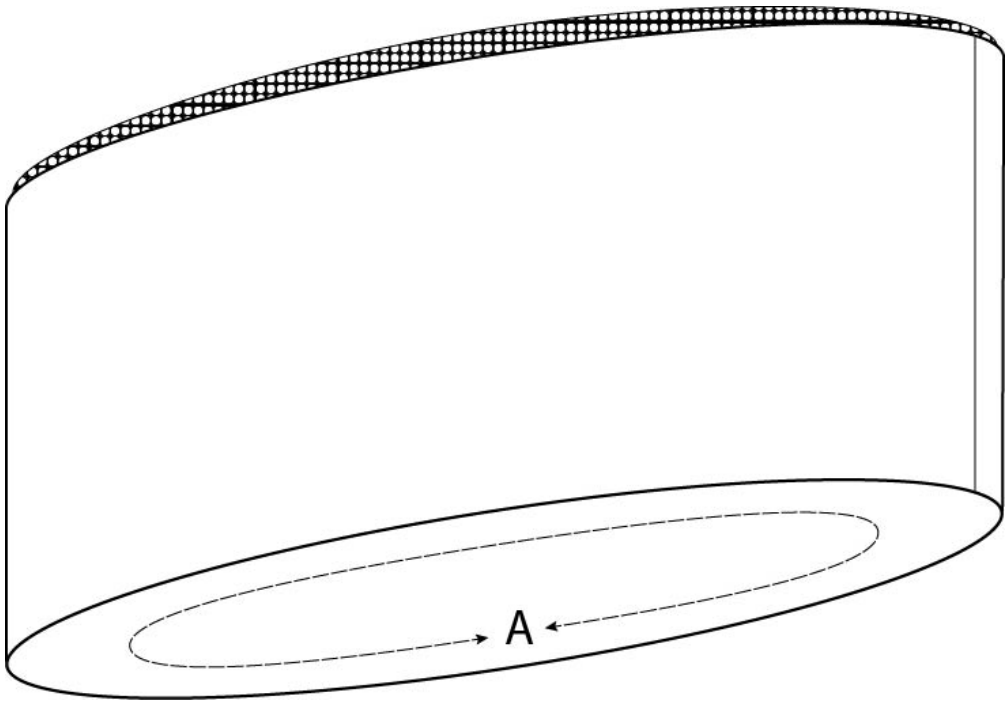


FIGURE 2. Inside – Food Handler's Cap

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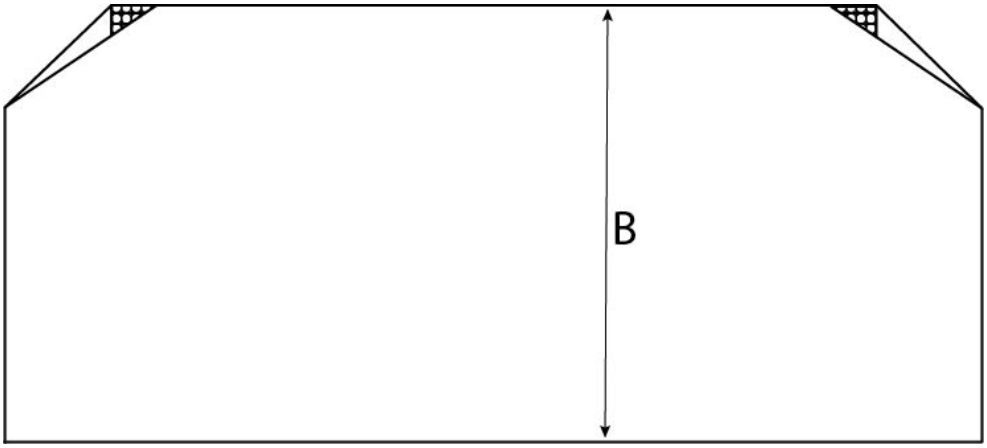


FIGURE 3. Flat – Food Handler's Cap

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